The Gamma-distribution with constant rate

Parametrisation

The Gamma-distribution has the following density

$$\pi(y) = \frac{b^a}{\Gamma(a)} y^{a-1} \exp(-by), \qquad a > 0, \quad b > 0, \quad y > 0,$$

where $\mathrm{E}(y) = \mu = a/b$ and $\mathrm{Var}(y) = 1/\tau = a/b^2$, where τ is the precision and μ is the mean.

This version of the Gamma-distribution is to make the rate b constant, so we use

$$\mu = \exp(\eta)$$

where η is the linear predictor, and

$$\tau = (s\phi)/\mu$$

where ϕ is the precision parameter (or $1/\phi$ is the dispersion parameter with this definition) and $s > 0$ is a fixed scaling (for the regression model), which gives Gamma-density with

$$a = s\phi\mu \qquad \text{and} \qquad b = s\phi$$

Link-function

The linear predictor η is linked to the mean μ using a default log-link

$$\mu = \exp(\eta)$$

Hyperparameter

The hyperparameter is the precision parameter ϕ, which is represented as

$$\phi = \exp(\theta)$$

and the prior is defined on θ.

Specification

- `family="gammasv"`.
- Required arguments: y and s (default value 1).

Hyperparameter spesification and default values

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hyper

theta

hyperid `58003`
name `precision parameter`
short.name `prec`
output.name `Precision-parameter for the Gammasv observations`
output.name.intern `Intern precision-parameter for the Gammasv observations`
initial `4.60517018598809`
fixed `FALSE`

prior `loggamma`
param `1 0.01`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

status `experimental`

survival `FALSE`

discrete `FALSE`

link `default log`

pdf `gammasv`

Example

In the following example we estimate the parameters in a simulated example.

```
n <- 300
phi <- 1.1
s <- runif(n, min = 0.8, max = 1.25)
x <- rnorm(n)
eta <- 1 + 0.2 * x
mu <- exp(eta)
a <- phi * s * mu
b <- phi * s
y <- rgamma(n, shape = a, rate = b)
r <- inla(y ~ 1 + x,
          family = "gammasv",
          scale = s,
          data = data.frame(y, x, s))
summary(r)
```

Notes

None.